

EMAD MAHMODI

Embedded Security & Firmware Analysis Researcher | Automotive / IoT Systems

emad.mahmodi.eng@gmail.com | +98 911 300 2150 | emad-mahmodi.github.io | Google Scholar: ZrJX2noAAAAJ

RESEARCH PROFILE

Computer Engineering researcher and R&D software lead with hands-on experience in firmware analysis, binary reverse engineering, embedded-system simulation, automotive security, protocol analysis, and production C++/Python development. Research record includes adaptive machine learning, concept drift, anomaly detection, and three peer-reviewed journal papers. Current interests focus on automated firmware analysis, security testing/fuzzing of embedded and IoT systems, program analysis, and AI-assisted binary understanding.

TARGET FIT: Firmware analysis • Embedded/IoT security • Reverse engineering • Protocol fuzzing • Program analysis • C/C++ • Python • Ghidra/IDA • AI/ML for security

PROFESSIONAL EXPERIENCE

R&D Team Lead / Senior Software Developer | Negar Khodro, Iran 2020–Present

- Lead R&D for automotive diagnostic and ECU-programming software, including production C++17/20, Qt/QML, Python tooling, protocol processing, automated analysis, validation, and performance engineering.
- Analyze PowerPC and TriCore ECU firmware and embedded communication behavior; work with UDS/KWP2000/CAN, binary-analysis workflows, reverse engineering, and security-focused system testing.
- Architect and develop ECUProg v2, a production multi-protocol programming suite used in 5,000+ automotive workshops, and coordinate technical development across complex hardware/software constraints.

Senior System Programmer | NIUniverse, Iran 2020–2022

- Performed binary security auditing, malware analysis, protocol simulation, vulnerability assessment, anti-debugging analysis, and investigation of virtual-function / v-table obfuscation in diagnostic software.
- Built C/C++ and Python analysis tools and reproducible workflows for dynamic behavior inspection, data extraction, and security testing.

Academic Researcher & Software Consultant | Independent / 2017–2020

Academic Collaboration

- Conducted machine-learning research in adaptive anomaly detection, concept drift, phishing detection, recommender systems, and quantitative model evaluation; contributed to peer-reviewed publications and experimental software.

EDUCATION

M.Sc. Computer Engineering – Software | Ferdowsi University of Mashhad, Iran | 2014–2017

GPA: 15.09/20 | Thesis: “Adaptive Rule-Base Phishing Detection Against URL Obfuscation” (18/20). Relevant graduate work: Statistical Pattern Recognition, Advanced Software Systems Security, Computer Networks Security, Parallel Algorithms, Expert Systems.

B.Sc. Computer Engineering – Software | Shomal University, Iran | 2008–2013

GPA: 14.02/20 | Final project grade: 20/20. Relevant coursework: Artificial Intelligence, Algorithms Design, Operational Research, Engineering Statistics and Probability, Data Structures, Networks, Operating Systems, Computer Architecture.

SELECTED RESEARCH & SECURITY PROJECTS

PISAD ECU Simulator & Firmware Security Testing

Developed a PowerPC-oriented ECU simulation environment for controlled firmware execution, register/memory inspection, breakpoint-driven analysis, analogue/CAN/UDS/KWP input injection, fault scenarios, and repeatable security experiments. The platform supports malformed diagnostic-frame injection and observation of firmware behavior for protocol-fuzzing research.

Security Auditing & Data Extraction of Android Diagnostic Systems

Reverse engineered Android-based automotive diagnostic applications and VCI communication. Used JADX/Frida, Bluetooth traffic capture, protocol inspection, and firmware/package extraction to study update logic, decrypted diagnostic data, and communication behavior across Android, VCI, and ECU layers.

LLM-Assisted Binary Code Understanding & Reverse Engineering

Built an evaluation/data-alignment workflow across source, assembly, and decompiled code using DWARF ground truth. Investigated function-name recovery, summarization, RAG-assisted firmware documentation, and robust benchmarking of code-focused language models.

Symbolic / Dynamic Analysis with SymCC & SymQEMU

Investigated issues and developed contributions around compiler-assisted and QEMU-based symbolic execution workflows, with emphasis on low-level program behavior, test-case generation, and binary-analysis tooling.

Locomotive GT26 Control System

Implemented compile-time/run-time state-machine logic in C++ and FPGA-to-PC interfaces using Verilog and shared memory for a railway control-system project, strengthening experience in real-time embedded logic and hardware/software integration.

PEER-REVIEWED PUBLICATIONS

E. Mahmodi, H. Sadoghi Yazdi, A. Ghaemi Bafghi (2020). "A drift aware adaptive method based on minimum uncertainty for anomaly detection in social networking." Expert Systems with Applications, 162, 113881. *First author; adaptive online learning, concept drift, uncertainty-aware model fusion.*

H. Koohi, Z. Kobti, T. Farzi, E. Mahmodi (2024). "UDIS: Enhancing Collaborative Filtering with Fusion of Dimensionality Reduction and Semantic Similarity." Electronics, 13(20), 4073. *Software, validation, data curation, visualization, and writing contributions.*

R. Barzegar Nozari, H. Koohi, E. Mahmodi (2020). "A Novel Trust Computation Method Based on User Ratings to Improve the Recommendation." International Journal of Engineering, 33(3), 377–386. *Recommender systems, clustering, quantitative evaluation.*

TECHNICAL SKILLS

Security & Analysis	Firmware analysis, binary reverse engineering, embedded/IoT security, program analysis, security testing, protocol fuzzing in controlled simulations, malware analysis
Programming	C++17/20, C, Python, Java, Verilog, Qt/QML
Reverse Engineering Tools	Ghidra, IDA Pro, Radare2, JADX, Frida, x32dbg; familiarity with SymCC/SymQEMU workflows
Embedded / Automotive	PowerPC, TriCore, CAN, UDS (ISO 14229), KWP2000 (ISO 14230), ECU firmware, hardware/software integration, FPGA interfaces
AI / Data	PyTorch, adaptive/online learning, concept drift, anomaly detection, clustering, dimensionality reduction, LLM evaluation, RAG
Research Practice	Experimental design, reproducible evaluation, data pipelines, benchmarking, scientific writing, peer-reviewed publication

RESEARCH DIRECTION

Interested in scalable methods for automated firmware analysis and fuzzing, vulnerability discovery in embedded/IoT software, program-analysis-guided security testing, and AI-assisted binary understanding. Particularly motivated by research that combines rigorous systems experimentation with deployable security tooling.